What do Sri Lankan crops need to grow well?

An analysis of the 40-crop cultivation requirements table, with simple explanations and charts

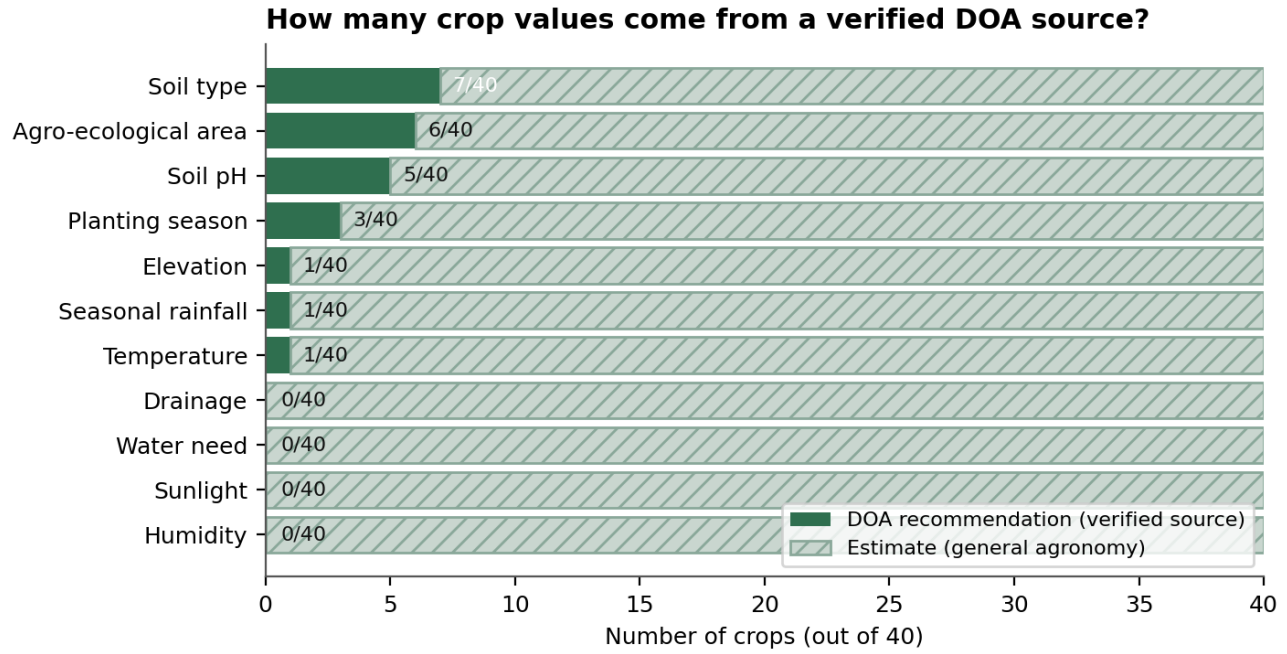
Key findings

- **The table is mostly estimates.** Only 1 of 40 crops has a temperature range from a verified DOA source, and 1 of 40 has a verified rainfall range. Humidity, sunlight, water need and drainage have none.
- **Temperature is the broad common ground.** The most crops (34 of 40) are happy at about 25°C. Only 3 of 40 still fit at 34°C, so heat is the first limit for the whole crop list.
- **Rainfall splits crops into two camps.** About 29 of 40 crops fit a season with roughly 1000 mm of rain, but only 8 fit at 2,000 mm.
- **Crops fall into 4 natural groups** by what they need (cool, acid-soil, warm-and-drier, warm-and-wetter). Group quality is moderate (silhouette 0.44).
- **Climate trends are not in this report.** Real NASA trend data could not be downloaded here, so none is shown. Page 8 explains how to produce it and how to read it.

> **How to use this report.** It describes patterns in the requirements table. It does not say what you should plant. Please check anything important with local agricultural officers and experienced farmers.

1. How reliable is the data?

The CSV has 40 crops and 30 columns. Each growing requirement has a status column that says whether the value comes from a **DOA recommendation** (an official Department of Agriculture source) or is an **estimate** from general agronomy. The chart counts how many crops fall in each case.



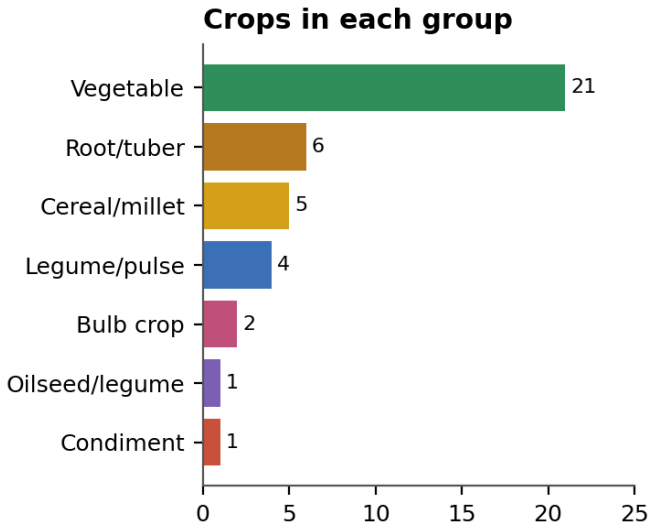
What the chart shows

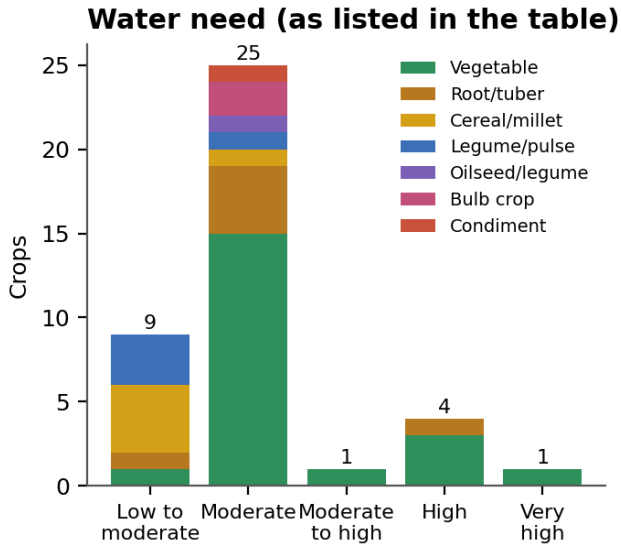
- Seven of the eleven requirement types have no more than 1 verified crop; humidity, sunlight, water need and drainage have none.
- Soil type (7 crops), agro-ecological area (6) and soil pH (5) are the best covered, but still cover under one-fifth of crops.
- Elevation is almost entirely qualitative (words such as low, mid or up-country). Only one crop has a number.

> **What this means.** Treat every range in this report as a starting point, not a proven limit. Hatched bars in later charts mark estimates, and solid bars mark verified values. The more hatching you see, the more local checking is needed.

2. What is in the table?

The 40 crops are mostly vegetables (21 crops), followed by root and tuber crops (6), cereals and millets (5), legumes (4), and a few bulb, oilseed and condiment crops. The right-hand chart shows how much water each crop is listed as needing.



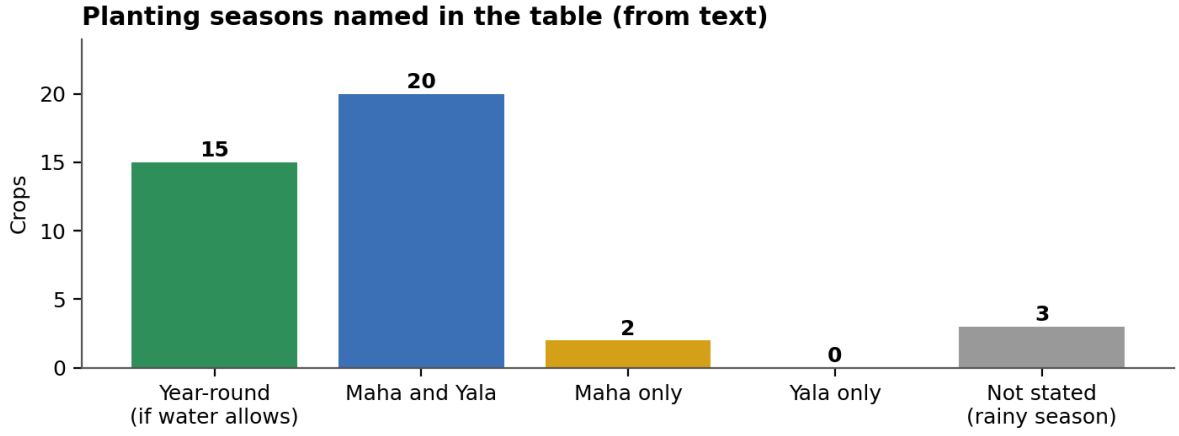


What the charts show

- Most crops (25 of 40) are listed as needing a **moderate** amount of water, and 9 more as low to moderate.
- Only 5 crops are listed as **high or very high** in water need: Ash Plantain, Gotukola, Mukunuwenna, Kiri Ala and Kangkung.
- The 5 crops listed as **low to moderate** and also having a seasonal rainfall maximum of 800 mm or less are Sorghum, Foxtail Millet, Green Gram, Cowpea and Black Gram.

Planting seasons

The planting season column is free text, so it was sorted by the words it contains. **Maha** is the main rainy season (about October to March) and **Yala** is the second season (about April to September).

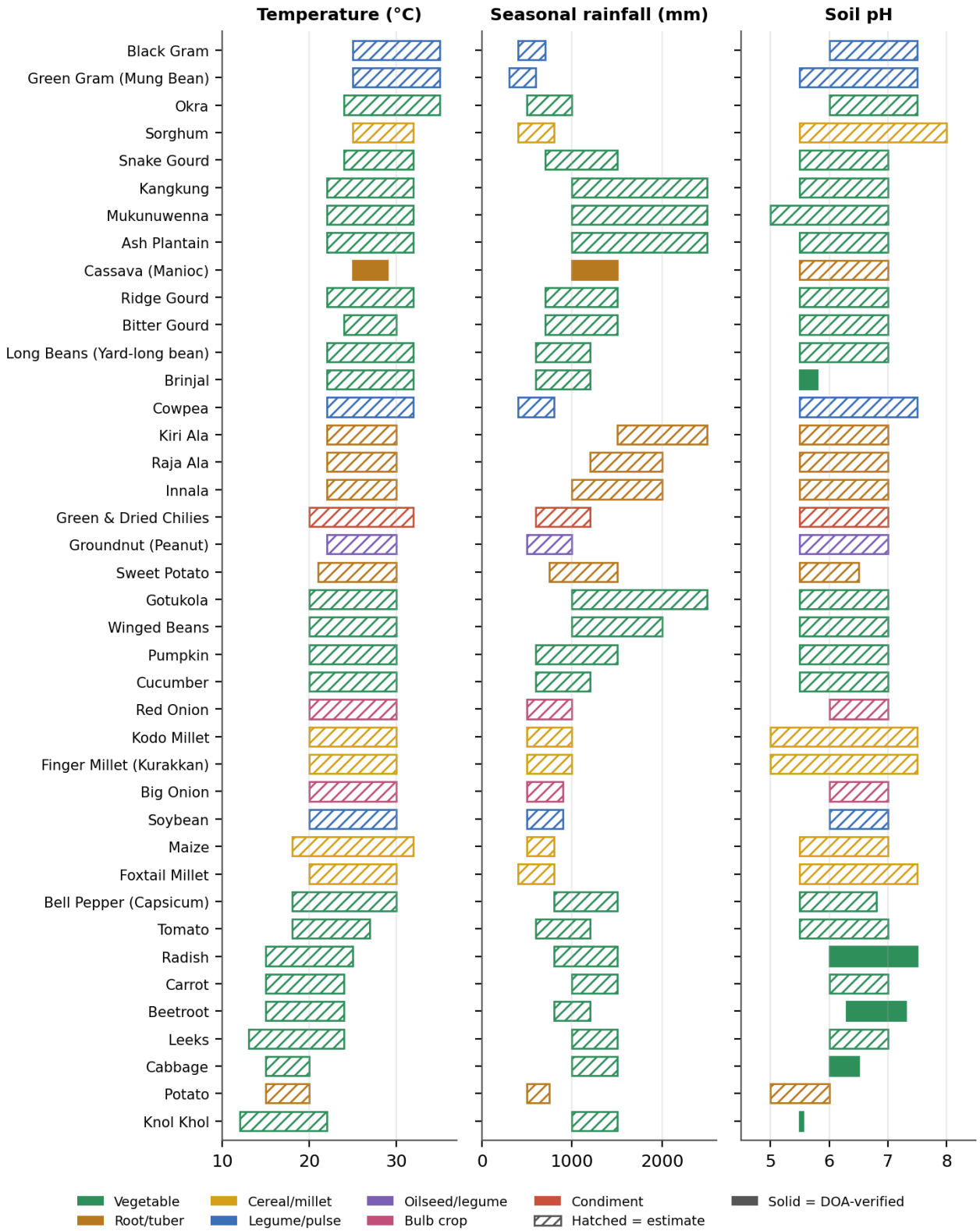


Fifteen crops are described as year-round, and 20 are named for both Maha and Yala. Three crops (Raja Ala, Foxtail Millet, Kodo Millet) only say rainy-season planting, with no Sri Lankan calendar given.

Note: the year-round count is the number of crops whose text says year-round. Some of these may still depend on irrigation, which the table does not describe.

3. What temperature, rainfall and pH do crops need?

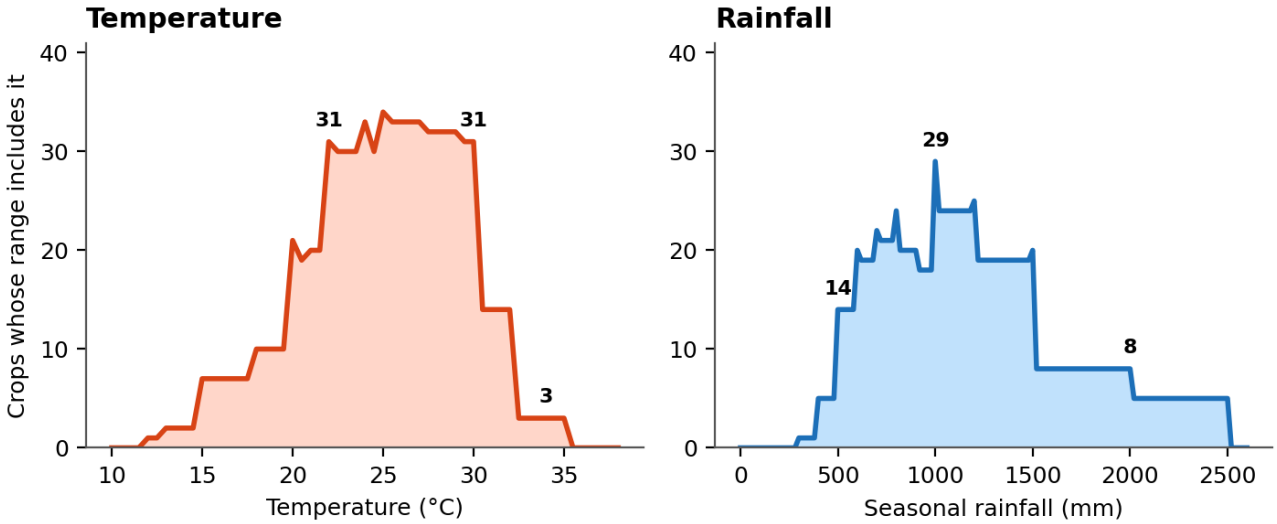
Each bar is the range a crop is listed as preferring. Crops are sorted from cool-loving (top) to warm-loving (bottom). Colour shows the crop group.



Only Cassava has a solid (verified) temperature and rainfall bar. Every other bar is an estimate. Potato's low rainfall range reflects the table's note that rainfall is managed through irrigation for that crop.

4. How many crops fit each condition?

For each temperature (left) and each seasonal rainfall (right), this chart counts how many of the 40 crops have that value inside their listed range. A high number means many crop choices. A low number means few.



What the charts show

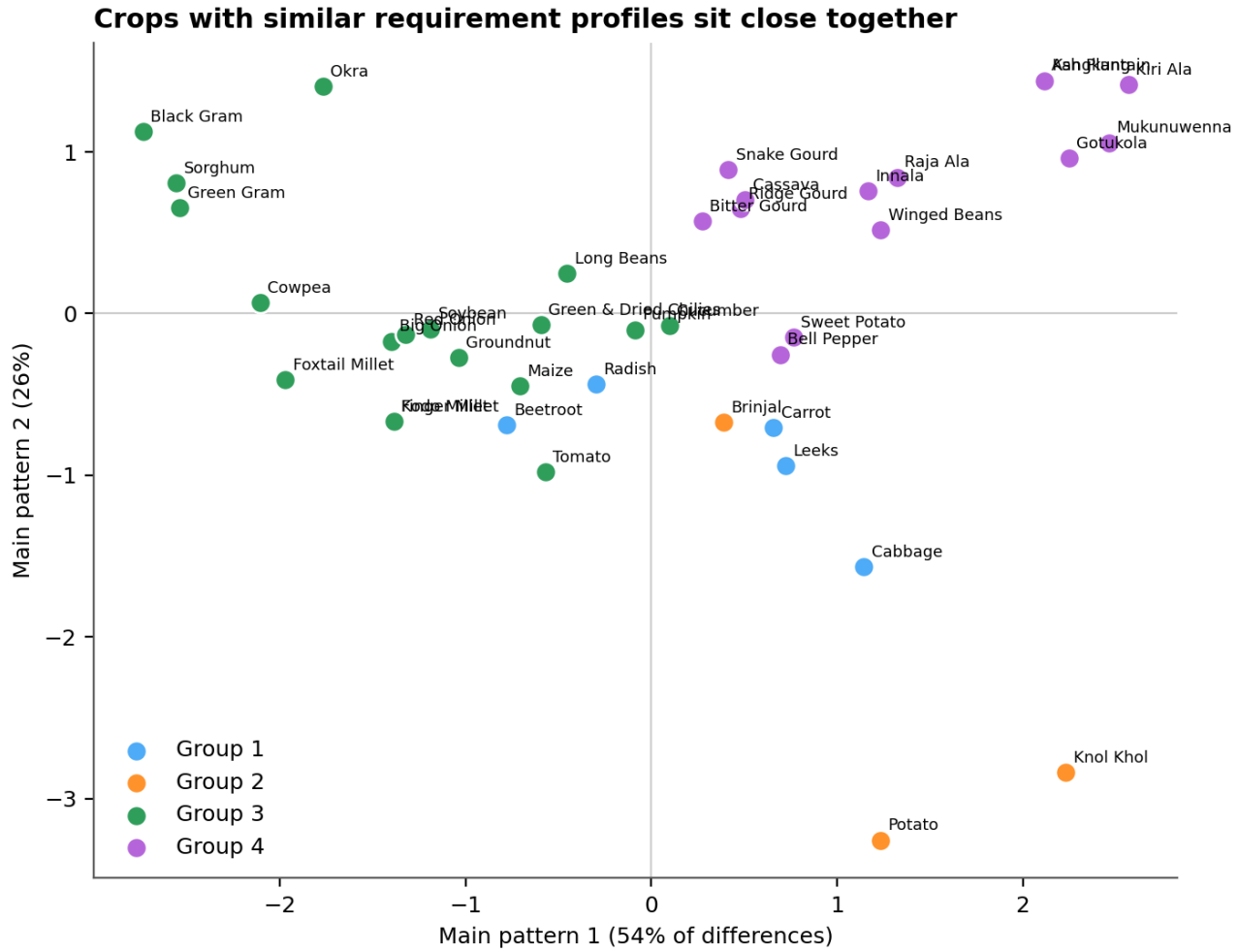
- **Temperature:** 31 crops fit at 22°C, 33 at 26°C and 31 at 30°C. This falls to 14 crops at 32°C and only 3 at 34°C. The peak is 34 crops at 25°C.
- **Cool crops are limited.** Only 10 crops fit at 18°C, and 7 crops (Carrot, Cabbage, Beetroot, Leeks, Radish, Knol Khol and Potato) have a top temperature of 25°C or less.
- **Rainfall:** 14 crops fit at 500 mm, 29 at 1,000 mm, 20 at 1,500 mm and only 8 at 2,000 mm. Choice is widest around 1,000 mm.

> **What this means.** If a location is hot (above 32°C) or very wet (above 2,000 mm), the number of suitable crops in the table drops sharply. In these places, crop choice and local advice matter most. Remember that the counts rest on mostly estimated ranges.

Limit of this view: it counts crops by one condition at a time. A crop also needs the right pH, soil, season and water, so the true list of suitable crops for a field is shorter.

5. Which crops are alike? (machine learning map)

To find crops with similar needs, each crop was described by the middle of its temperature, rainfall, pH and humidity ranges. A method called **PCA** squeezes these four numbers onto a flat map, so similar crops land close together. A method called **K-means** then groups nearby crops. No crop yields were used, because the table has none.



Group	Crops	Avg temp	Avg rain	Avg pH	Avg humidity	What they share
1	5	19°C	1180 mm	6.6	68%	Cool-season vegetables
2	3	20°C	925 mm	5.5	68%	Cool-ish, acid-soil crops
3	18	26°C	736 mm	6.4	60%	Warm and drier: millets, pulses, onions
4	14	26°C	1459 mm	6.2	74%	Warm and wetter: gourds, leafy greens, roots

The two map directions explain 54% and 26% of the differences between crops (80% together). Left to right mostly separates drier crops from wetter, more humid ones. Bottom to top mostly separates cooler crops from warmer ones. The grouping score (silhouette 0.44) is moderate: the groups are real but their edges are blurry.

These groups describe the **table's numbers**, many of them estimates. They are not proof that crops in a group behave alike in the field.

6. Simple ways to use these results

- **Start with the fixed limits.** Check your soil pH and whether the field is wet or well drained. These cannot be changed easily.
- **Then check temperature and rainfall** for your area and season using Chart 4 and the range chart on page 4. Look for crops whose bars include your values.
- **Prefer verified (solid) values** when they exist, and treat hatched estimates as questions to ask a local expert.
- **Use the groups to find alternatives.** If a crop is not suitable, other crops in the same group have similar needs and may be worth checking.
- **Remember the missing facts.** The table has no yields, no prices, no pest history and no nitrogen-fixing or rooting-depth information, so it cannot rank crop rotations by itself.

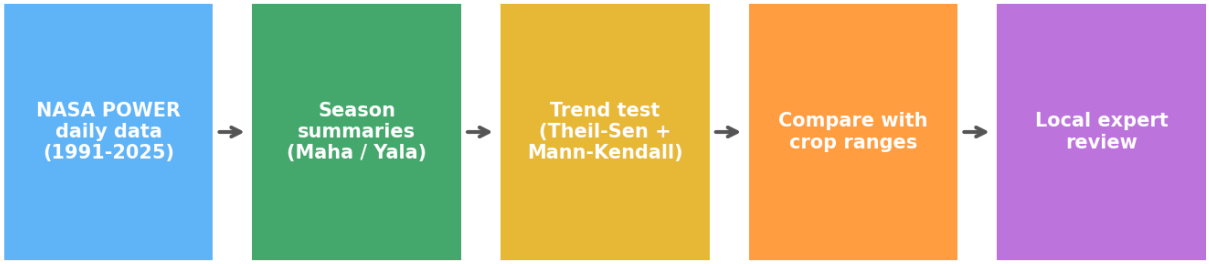
What this report cannot tell you

- Whether a crop will actually do well in a specific field.
- How climate is changing in your area (needs NASA data; see page 8).
- Which crop earns the most money or best protects the soil.

> **Important.** This analysis supports decisions. It does not replace farmer judgment or local agricultural expertise.

7. Climate trends: how to get them (not included here)

Trend charts must come from real measurements. NASA's POWER data service could not be reached when this report was made, so **no climate trend or forecast is shown**. Showing invented numbers would be misleading. Your two Colab notebooks do this job; the steps are below.



How the trend analysis works

- **Season summaries:** for each year, the mean temperature, total rainfall and mean humidity of the Maha (or Yala) season.
- **Theil-Sen slope:** the typical change per year, calculated in a way that is not thrown off by one extreme year.
- **Mann-Kendall test (p-value):** asks whether a trend is real or could be chance. A p-value below 0.05 means a trend is detectable. Above 0.05 means **no reliable trend**, so do not rely on the extended line.
- **Back-testing:** a forecast is tested on months it has not seen and compared with a simple average. If it does not beat the average, the average is used.

Steps to finish this analysis

- Open **crop_trends_ml_colab.ipynb** in Google Colab, set your location and season, and run all cells.
- Check the trend verdict on each panel before trusting any extended line.
- Share the numbers or the notebook output with me and I can add real trend charts to this report.

NASA POWER values are gridded (about 0.5°) and come from satellite and model data. They describe the surrounding area, not one field, so compare them with local rain-gauge records.